

		Course Name Power Plant Electrical Equipment (MPA-ELE 25/26Z)		
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Task Number B	Task Name Measurements on a stabilized differential protection SPCD 3D53			

Assignment

The objectives of this laboratory exercise were:

1. **Calculate Settings:** To calculate the setting module parameters for the SPCD 3D53 Differential Protection for a specified power transformer.
2. **Test Characteristics:** To test the tripping characteristic in all three area settings (Section 1: Basic sensitivity, Section 2: First slope, Section 3: Second slope) and compare the measured values with the protection settings.
3. **Visualization:** To calculate the stabilizing current (I_b) and differential current (I_d) from the measured phase currents (I_1 I_2) and plot the operating characteristic.

Theoretical introduction

Stabilized Differential Protection The protection relay is designed to protect two-winding power transformers and generators against inter-phase faults and inter-turn short circuits.

Principle of Operation:

The operation is based on the comparison of currents entering and leaving the protected zone.

1. **Differential Current (I_d):** This is the difference between the primary (I_1) and secondary (I_2) currents. It represents the fault current.

$$I_d = |I_1 - I_2|$$

2. **Stabilizing Current (I_b):** To prevent false tripping during external faults (which cause CT saturation), a stabilizing current is calculated representing the average load.

$$I_b = \frac{|I_1| + |I_2|}{2}$$

Procedure:

The lab work focused on verifying the tripping characteristic of the relay using secondary injection.

Setup and Wiring:

The protection relay was configured using **Omicron CMC 256 Plus** software. To perform the measurements, an **Omicron 363** advanced protection test set was utilized. This device is capable of generating six independent current outputs, allowing for the simultaneous simulation of both the primary (HV) and secondary (LV) sides of a power transformer.

We connected a total of 6 current points to the measuring inputs of the SPCD 3D53 relay (Terminal Block X₂). The wiring configuration was established as follows:

- **HV Side Simulation (I_1):** Three current outputs from the Omicron were connected to the High Voltage measuring inputs (L_1, L_2, L_3) of the relay.
- **LV Side Simulation (I_2):** The other three current outputs were connected to the Low Voltage measuring inputs (L_1, L_2, L_3).

Wiring Schema:

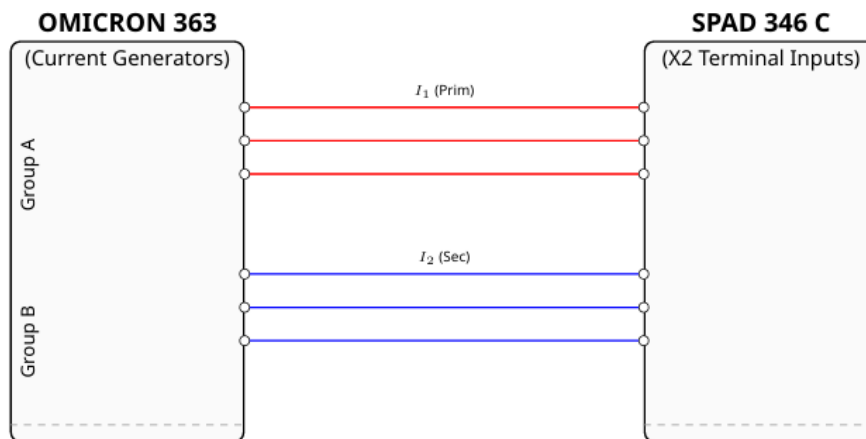


Figure 1: Wiring schematic: 6-point current injection from Omicron 363 to SPAD 346 C measuring inputs.

This setup enabled the verification of the differential protection logic under balanced and unbalanced three-phase conditions.

Measurement Methodology:

The nominal current (I_n) was set to 1 A. We recorded the input currents (I_1 I_2) at the tripping point and calculated I_1 and I_2 .

To verify the operating characteristic, we performed measurements in all three defined sections of the curve. The measurement procedure was consistent for each step:

1. We fixed the value of the primary current I_1 (e.g., 150 mA).
2. We gradually increased the secondary current I_2 (starting from a balanced state or zero) until the differential protection tripped.
3. We recorded the values of I_1 and I_2 at the exact moment of tripping.
4. This process was repeated for two measurement points in each of the three sections: (Sensitivity, Slope 1, and Slope 2) to accurately map the characteristic curve.

Measurement Results:

The nominal current (I_n) was set to 1 A. We recorded the input currents (I_1 I_2) at the tripping point and calculated I_b and I_d .

Table 1: Measured and Calculated Values (All values in mA)

Section	Meas. #	I_1 (mA)	I_2 (mA)	Calculated I_b	Calculated I_d
I (Sensitivity)	1	150	310	230	160
	2	300	460	380	160
II (Slope 1)	3	800	1050	925	250
	4	1300	1660	1480	360
III (Slope 2)	5	2800	6130	4465	3330
	6	3300	7630	5465	4330

Calculations Sample:

For Measurement 1 (Section 1):

$$I_b = \frac{150 + 310}{2} = 230 \text{ mA} \quad ; \quad I_d = |150 - 310| = 160 \text{ mA}$$

For Measurement 5 (Section 3):

$$I_b = \frac{2800 + 6130}{2} = 4465 \text{ mA} \quad ; \quad I_d = |2800 - 6130| = 3330 \text{ mA}$$

Characteristics Plot:

The following graph visualizes the operating characteristic based on our measurements. The plotted points (Blue) represent our measured values, outlining the three distinct sections of the protection curve.

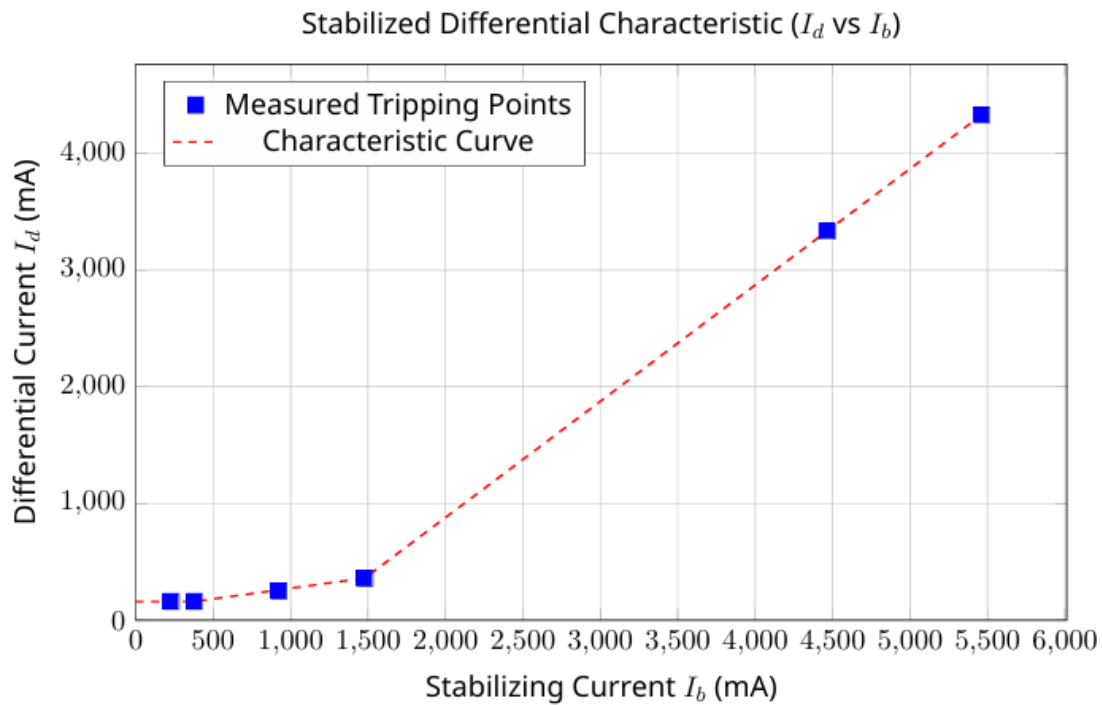


Figure 2: Plot of Measured Stabilized Differential Characteristic

List of Instruments:

1. **Protection Relay Configuration:** SPCD 3D53
2. **Protection Test Set:** Omicron CMC 256 Plus (used for secondary injection)
3. **Software:** Omicron Test Universe (for plotting currents and testing)
4. **Connection Cables:** Standard lab banana cables

Conclusion:

The laboratory exercise successfully demonstrated the operation of the differential protection. The measurements confirmed the three distinct sections of the operating characteristic:

- In **Section 1**, with I_1 fixed at low values, the tripping occurred at a constant differential current of 160 mA, verifying the basic sensitivity.

- In **Sections 2 and 3**, as we increased the currents, the differential current required to trip the relay increased. This confirmed the proper stabilized operation set via SPCD 3D53.

The use of the Omicron CMC 256 Plus and its software allowed for precise visualization and verification of the protection logic.